

MAXWELL SHEPHERD

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EXPERIENCE

Boeing

Data Scientist, Industrial Engineering

August 2023 - Present

- Pioneered the organization's first agentic AI / computer-use automation systems, shipping production agents to 6 teams that save an estimated 400 hours per week of manual workflows.
- Cut schedule flow times 56% by delivering ML and statistical models for aircraft asset management on multi-terabyte datasets, shaping engineering resource allocation for 480 employees.
- Evaluated vision transformers for dense prediction on in-domain engineering drawings under limited labeled data.
- Shipped a self-serve analytics platform (Python, SQL, React) for enterprise-scale metrics; adopted by 1,100 employees, 260 daily users.
- Cut ETL latency from 24 hours to 15 minutes by re-architecting pipelines that process 450M+ rows daily.
- Authored a data synthesis and retrieval suite used 21,000 times per month by 1,000+ employees, credited with \$2.2M yearly cost savings.

Boeing Designated Expert, Analytics

June 2025 - Present

- Recognized as enterprise-wide authority on business statistics, machine learning/GenAI, and analytics for a 2,000+ person organization.
- Responsible for setting technical direction for future technologies and airplane programs.

Industrial Engineer (Intern, I, II)

May 2019 - August 2023

Disney, Engineering Intern

May 2018 - August 2018

PTC Inc., Engineering Intern

May 2016 - August 2016

RESEARCH & PUBLICATIONS

- "A benchmark of expert-level academic questions to assess AI capabilities," *Nature* (2026). Contributing author via the HLE Contributors Consortium.
- "Frozen Vision Transformers for Dense Prediction on Small Datasets: A Case Study in Arrow Localization," *arXiv:2604.16758* (2026).
- "Concurrent Streaming, Viewer Transfers, and Audience Loyalty in a Creator Ecosystem: A Minute-Level Longitudinal Study," *arXiv:2603.23773* (2026).

TECHNICAL SKILLS

Languages: Python (async, concurrency), SQL, JavaScript, R, MATLAB

ML & GenAI: PyTorch, XGBoost, Claude/OpenAI Agent SDKs, MCP, Claude Code, Codex

Data & Infrastructure: PostgreSQL, InfluxDB, Schema Design, Indexing & Query Optimization, Partitioning, Docker, Airflow, Linux, Git

Web & Visualization: Flask, FastAPI, React (HTML/CSS/JS, Material-UI), Tableau, Grafana

EDUCATION

Johns Hopkins University

August 2024 - May 2026

M.S. Data Science with Honors

GPA: 4.00

- Specialization in Machine Learning & Artificial Intelligence
- Coursework: Applied Generative AI, Computational Statistics, Human-Computer Interaction, Big Data Engineering, Distributed Computing

Purdue University

August 2016 - May 2020

B.S. Industrial Engineering

GPA: 3.53

- Dean's List & Semester Honors
- Coursework: Operations Research, Optimization, Stochastic Modeling, Manufacturing Processes